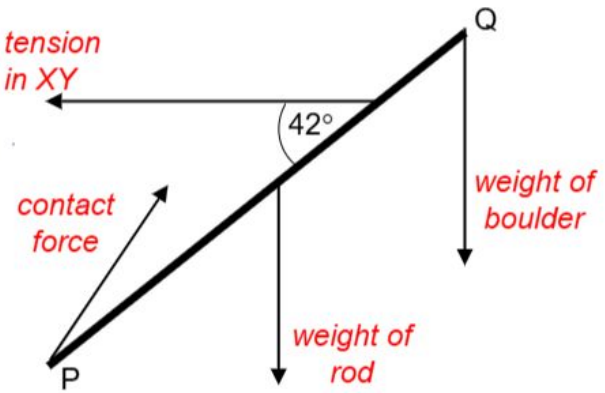
2(a)	**Resultant force** acting on the rod must be **zero** (*translational equilibrium*). **Resultant torque / moment** of the rod **about any point** must be **zero** (*rotational equilibrium*).	**B1** **B1**
(b)	 1 mark for every 2 forces correctly drawn and labelled.	**A1** **A1**
(c)	Using principle of moments, taking moments about P, $(W_{boulder})(6.0 \cos 42°)+(W_{rod})(6.0/2 \cos 42°) = (T_{XY})(2.2)$ $17g\ (6.0 \cos 42°) + 15g\ (6.0/2 \cos 42°) = (T_{XY})(2.2)$ $T_{XY} = 487$ N $= \underline{490\ \text{N}}$ (2 sf)	**C1** **A1**
(d)	T_{XY} will **increase**. When the mass of the 'load' increases, the <u>sum of clockwise moments</u> due to weight of rod and weight of crate will <u>increase</u>, resulting in the need for <u>anticlockwise moments</u> due to T_{XY} to <u>increase</u> as well.	**A1** **M1**